1. First, lock down exactly what your thesis is about

Final core problem

Your system receives:

AIS data
→ where vessels say they are, when they broadcast, speed, course, MMSI, etc.

SAR data
→ where vessels are physically detected in satellite radar imagery.

Then your system asks:

> **"Does this SAR-detected vessel have a consistent AIS counterpart at approximately the same location and time?"**

From that, you identify:

- matched vessel
- missing AIS
- inconsistent AIS
- suspicious discrepancy

The thesis is therefore fundamentally about:

AIS–SAR cross-modal vessel matching + anomaly/discrepancy detection.

This is much safer than saying:

> SAR detection = illegal vessel
> Dark vessel = illegal dumping

because those conclusions do not logically follow.

Global Fishing Watch itself distinguishes SAR detections from AIS matching and notes that unmatched SAR detections may have several explanations; it does not treat every unmatched vessel as proof of illegal activity.

---

2. Your title

The dean's suggested title is already strong:

> **A Cross-Modal Spatiotemporal Graph Framework for Dark-Vessel Detection Using AIS Trajectories and SAR Imagery**

I would keep this for now.

You can later decide whether **“dark-vessel detection”** or **“AIS discrepancy detection”** is the more scientifically precise wording after you establish your labeling methodology.

A very important point:

Define “dark vessel” operationally.

Do **not** define it simply as:

> “A vessel that does not have AIS.”

Instead, define a measurable rule such as:

> A SAR-detected vessel is considered an AIS-dark candidate when no sufficiently reliable AIS trajectory can be matched to the SAR detection within a predefined spatial and temporal tolerance.

Then you must define the tolerance.

For example:

SAR timestamp: 10:15:00
Search AIS: 10:00–10:30
Spatial tolerance: X meters/km

Those numbers should be justified experimentally or from literature/data characteristics—not chosen arbitrarily.

---

3. Your most important research question

Your thesis should revolve around something like:

> **Can a cross-modal spatiotemporal model improve detection of AIS–SAR discrepancies compared with single-modality and rule-based approaches in a congested maritime environment?**

Then supporting questions can address:

1. How accurately can AIS trajectories and SAR vessel detections be aligned?
2. How effectively can the proposed model identify unmatched/inconsistent vessels?
3. Can the model reduce false positives caused by congestion, slow movement, loitering, and AIS transmission gaps?
4. Does the proposed cross-modal consistency mechanism actually improve performance?
5. What is the computational cost and inference latency of the approach?

That gives your thesis a very clean story.

---

4. Your biggest feasibility checkpoint: DATA

This is the part I would investigate **before spending too much time designing the neural network**.

Your thesis depends on having:

> **AIS + SAR + time + location overlap**

for the same study area.

Sentinel-1 data are available through the Copernicus Data Space Ecosystem, including historical GRD products from October 2014 onward, and Sentinel-1 provides radar imaging that works day/night and through cloud cover.

Global Fishing Watch also provides AIS-related APIs and SAR vessel detections, including a SAR dataset containing industrial vessels detected through satellite imagery. Its current API documentation identifies the SAR dataset as `public-global-sar-presence:latest` and includes whether a detection was AIS-matched.

This creates an important decision for your team:

Option A — Raw Sentinel-1 imagery

You obtain:

AIS trajectories + raw Sentinel-1 SAR images

Then your team performs:

SAR preprocessing
→ vessel detection
→ vessel position extraction
→ AIS matching
→ graph construction
→ anomaly detection

Advantage

Much stronger claim of building a genuine multimodal system.

Disadvantage

Much more work.

Option B — GFW SAR vessel detections

Instead of processing raw SAR imagery yourself, use already-derived SAR vessel detections.

GFW provides SAR vessel detections derived from Sentinel-1 and makes them available through its APIs/data products.

Then:

AIS trajectories + SAR vessel detection coordinates/features

→ cross-modal matching/anomaly detection.

Advantage

Much easier and more feasible for an undergraduate thesis.

Disadvantage

Your "SAR modality" is no longer really the raw raster-image modality described in your original idea.

---

5. I would NOT decide the dataset casually

You should make a **Data Feasibility Matrix** early.

Requirement	Must have?
Batangas Bay coverage	✅
Exact SAR acquisition timestamp	✅
SAR vessel locations	✅
AIS vessel positions	✅
AIS timestamps	✅
Vessel identifiers	Preferably ✅
Historical data	✅
Spatial coordinates	✅
Sufficient overlapping dates	✅
Congested/normal vessel traffic	✅
AIS gaps	✅
SAR unmatched detections	✅
Data licensing permits thesis use	✅

Your team should literally test this before model development.

---

6. Don't assume MarineTraffic will be available

Your original proposal says:

> Global Fishing Watch or MarineTraffic API

Don't leave this vague.

You need to establish exactly:

- what data source you'll use
- how you obtain it
- what historical period is available
- whether API access is available to students
- usage/licensing restrictions
- spatial resolution
- temporal resolution
- whether individual AIS messages are accessible

GFW currently requires registration/API access and states that its APIs are available for non-commercial purposes, which may fit academic research, but you still need to follow its attribution/usage requirements.

Recommendation for your proposal

Don't write:

> “AIS data will be obtained from Global Fishing Watch or MarineTraffic.”

Write:

> “AIS data will be obtained from [specific confirmed source], subject to its available historical coverage and data-use terms.”

Confirm this before finalizing the methodology.

---

7. Define your study area

Don't say:

> “Batangas Bay”

and leave it there.

You need a geographical boundary.

For example:

Study Region

- defined polygon
- latitude/longitude bounds
- port area
- surrounding navigational waters
- exclusion zones if necessary

You could have:

Core Area

Batangas Port and immediate surrounding waters.

Extended Area

A surrounding buffer to capture incoming/outgoing vessels.

This helps prevent the thesis from becoming enormous.

The dean explicitly recommended using a **limited geographic area and selected periods**, which is a very good idea for feasibility.

---

8. Define your study period

Do NOT simply say:

> 2024–2026

Instead determine:

- number of SAR scenes
- acquisition dates
- overlapping AIS periods
- seasonal conditions
- weekdays/weekends if relevant
- congested periods
- relatively quiet periods

You could eventually construct something like:

Training: January–June
Validation: July–August
Testing: September–October

But this must be decided based on actual data availability.

---

9. VERY IMPORTANT: define your ground truth

This is probably the **hardest research-methodology issue** in your thesis.

How do you know that something is actually a dark vessel?

You cannot simply say:

> SAR detection + no AIS = ground-truth dark vessel.

Because the AIS could be missing because of:

- receiver coverage
- transmission interval
- signal interference
- data processing limitations
- incorrect vessel association
- geolocation differences
- timestamp mismatch
- vessels not required to carry AIS
- satellite acquisition timing

GFW itself documents AIS coverage gaps and notes that AIS transmissions can vary in frequency and completeness.

So your thesis needs an **objective labeling protocol**.

---

10. Create a Dark-Vessel Labeling Protocol

For every SAR detection:

Step 1 — Find nearby AIS transmissions

Search within:

Δt = predefined temporal window

and:

Δd = predefined spatial radius

Step 2 — Check candidate vessels

Compare:

- distance
- time
- heading
- speed
- trajectory continuity
- vessel dimensions if available

Step 3 — Assign a label

Something like:

Class 0 — Matched

Reliable AIS counterpart exists.

Class 1 — AIS discrepancy

SAR vessel detected but AIS counterpart is unavailable/inconsistent under your matching criteria.

Class 2 — Ambiguous

Potential counterpart exists but evidence is insufficient.

For evaluation, you might exclude ambiguous cases from the primary classification metric rather than pretending they are definitely positive or negative.

That makes your methodology much more defensible.

---

11. Separate "dark" from "anomalous behavior"

This is another major issue.

A vessel can be:

AIS-dark but behaving normally.

A vessel can also be:

AIS-visible but behaving unusually.

Those are different phenomena.

Therefore your thesis should have two concepts:

AIS discrepancy

> Does the observed SAR vessel correspond to a reliable AIS transmission?

Behavioral anomaly

> Given the vessel's movement/context, does its behavior differ from expected maritime patterns?

Your dean specifically wants you to account for:

> **normal loitering and slow movement**

So don't let your model learn:

> "slow = suspicious."

---

12. Build a congestion-aware representation

For each vessel, consider features such as:

AIS features

- latitude
- longitude
- timestamp
- speed over ground
- course over ground
- heading
- rate of turn if available
- acceleration
- trajectory curvature
- transmission interval
- AIS gap duration

SAR features

- detection latitude
- detection longitude
- timestamp
- estimated vessel size
- detection confidence if available
- orientation
- radar-related characteristics available from your chosen dataset

Context features

- distance to port
- nearby vessel density
- traffic lane density
- local congestion
- distance to shoreline
- vessel type if available
- time of day

You don't necessarily need all of these.

The goal is to distinguish:

> **“AIS disappeared because the environment/data made matching difficult”**

from

> **“This SAR observation is genuinely inconsistent with the available AIS trajectory.”**

---

13. Decide what your graph actually represents

This needs to be crystal clear in Chapter 3.

Your graph could be:

Nodes

Each node = vessel observation at a specific time.

For example:

```
vessel A @ 10:00
```

```
vessel A @ 10:05
```

```
vessel B @ 10:05
```

Edges

Edges can represent:

- temporal continuity
- spatial proximity
- vessel trajectory connection
- neighborhood interaction
- AIS-to-SAR candidate correspondence

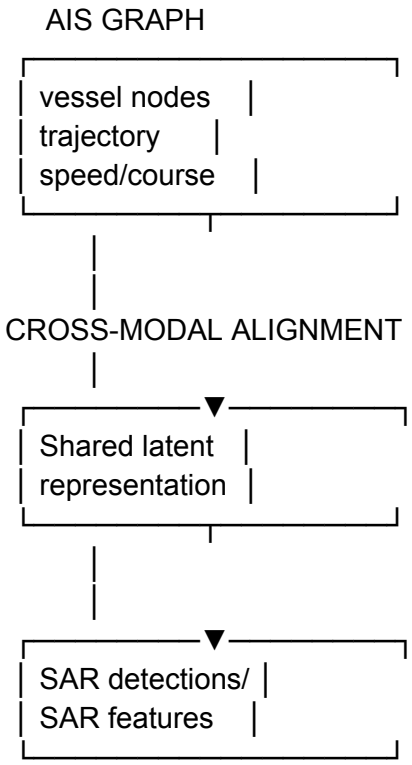
This is where your GNN becomes meaningful rather than simply being a buzzword.

---

14. You need a cross-modal graph design

Your thesis's most important technical contribution should probably live here.

Conceptually:



The key question becomes:

> **How does your model decide that an AIS observation and a SAR observation correspond to one another?**

That is exactly what your dean is asking you to explain.

---

15. Your proposed cross-modal consistency mechanism

You need to mathematically define it.

For example, you could have:

AIS encoder

$z_{AIS}=f_{AIS}(G_{AIS})$

SAR encoder

$z_{SAR}=f_{SAR}(X_{SAR})$

Alignment

sij=sim(zAISi,zSARj)s_{ij}=sim(z_{AIS_i},z_{SAR_j})

where `sim()` measures compatibility.

Then:

LalignL_{align}

penalizes AIS/SAR representations that should correspond but are far apart.

Your final objective could have something conceptually like:

L=Lreconstruction+λLalignment+γLconsistencyL = L_{reconstruction}+\lambda L_{alignment}+\gamma L_{consistency}

You don't need to commit to this exact formulation yet, but your thesis needs this level of specificity eventually.

---

16. Be careful with the phrase "novel architecture"

Your dean specifically said:

> AIS-SAR fusion already exists.

Therefore your novelty cannot simply be:

> “We used AIS + SAR + GNN.”

That's not enough.

Your novelty needs to be something like:

> **A specific cross-modal spatiotemporal consistency/alignment mechanism designed for AIS–SAR discrepancy detection under congested maritime conditions.**

Then you need to prove:

> removing that component makes the system worse.

Which leads directly to your ablation study.

---

17. Your ablation study is mandatory

You should have:

Model A

Full proposed model

Model B

Without cross-modal alignment

Model C

Without contrastive/consistency component

Possibly:

Model D

Without congestion/context features

Then compare:

F1

Precision

Recall

false-positive rate

etc.

If your special component does not improve anything, that's an important scientific result too.

---

18. Baselines

Your dean explicitly requested manageable baselines.

I would organize them like this:

Baseline 1 — Rule-based AIS-SAR matching

Example concept:

> nearest AIS vessel within spatial + temporal threshold.

Very important because it gives you a simple conventional benchmark.

Baseline 2 — AIS-only anomaly detector

Only AIS information.

Baseline 3 — SAR-only method

Only SAR information.

Baseline 4 — Proposed cross-modal model

AIS + SAR + your alignment mechanism.

This gives you the central comparison:

single modality vs simple matching vs proposed multimodal method.

---

19. Be careful with U-Net

Your original proposal says:

> compare against standard U-Net architectures.

That may not be the most appropriate baseline for your **final task**.

U-Net is primarily a segmentation architecture.

Your core task is:

> **cross-modal vessel matching/discrepancy detection.**

So comparing your final anomaly detector directly with U-Net may not be apples-to-apples unless you're explicitly doing SAR vessel segmentation/detection as a subtask.

I'd only include U-Net if your experimental pipeline contains:

> SAR image → vessel segmentation/detection.

Otherwise, a matching/anomaly baseline makes much more methodological sense.

---

20. Evaluation metrics

Your dean already identified the important ones.

Classification

Precision=TPTP+FPPrecision = \frac{TP}{TP+FP} Recall=TPTP+FNRecall = \frac{TP}{TP+FN} F1=2PRP+RF1 = \frac{2PR}{P+R}

False-positive rate

Especially important because Batangas is congested.

FPR=FPFP+TNFPR=\frac{FP}{FP+TN}

Matching accuracy

How often does the system correctly associate:

SAR detection ↔ AIS vessel

Detection latency

How long does the system take to process a case.

Computational efficiency

Consider:

- inference time
- GPU/CPU requirements
- memory consumption
- number of parameters
- throughput

You don't need to measure everything. Pick metrics that directly support the research questions.

---

21. Congestion-specific evaluation

This should be one of your strongest experiments.

Divide test cases into something like:

Low congestion

Few nearby vessels.

Medium congestion

Moderate traffic.

High congestion

Many nearby vessels.

Then measure whether false positives increase.

That directly addresses the dean's concern:

> **Can the system distinguish a real discrepancy from ordinary port congestion?**

This could become one of the most compelling parts of your thesis.

---

22. Also evaluate normal loitering

Create cases where vessels:

- slow down
- stop temporarily
- anchor
- turn repeatedly
- travel slowly around port
- remain near other vessels

Your system should **not automatically classify these as dark-vessel events**.

This is important because your model shouldn't accidentally learn:

> unusual movement = missing AIS.

---

23. Data leakage prevention

This is extremely important for trajectory data.

Imagine:

Training
Vessel A, January 1–10

Testing
Vessel A, January 11

Your model may effectively see the same vessel's behavior during training and testing.

That can inflate performance.

Your dean specifically warned about this.

Better approach

Separate data by:

- vessel
- time
- geographic region where appropriate

For example:

Train: vessels/time period A
Validation: separate vessel/time period
Test: unseen vessel/time period

At minimum, document exactly what unit you use to separate the splits.

---

24. Avoid random row splitting

This is especially important.

Don't just do:

80% rows = training
20% rows = testing

because neighboring trajectory points from the same voyage could appear in both sets.

Instead, split using:

voyages / vessel IDs / time blocks / scenes

depending on the exact experimental design.

---

25. SAR preprocessing

If using raw Sentinel-1 imagery, you'll need to document your preprocessing pipeline.

Potential stages include:

Sentinel-1 GRD
↓
Scene selection
↓
Subset Batangas Bay
↓
Radiometric calibration
↓
Noise/border handling
↓
Speckle treatment if justified
↓
Land/sea masking
↓
Normalization
↓
SAR chips
↓

Vessel detection/features

Sentinel-1 GRD products are already detected and projected to ground range, and the Copernicus documentation describes processing options including backscatter products.

Do not invent preprocessing steps just because they sound standard. Your final pipeline should depend on the actual product you're using.

---

26. SAR vessel detection itself must be resolved

This is an important scope decision.

There are actually **two different ML problems** hiding inside your thesis:

Problem A

Detect vessel from SAR image.

Problem B

Determine whether SAR vessel corresponds to AIS.

If your team tries to solve both from scratch, your project could become huge.

You therefore need to decide:

Route 1

Use an existing SAR vessel detector/dataset and focus your novelty on AIS–SAR alignment.

Route 2

Build your own SAR vessel detector and then perform alignment.

For a 3-person undergraduate thesis, I'd strongly consider **Route 1 unless your adviser explicitly requires raw-image vessel detection**.

GFW already provides Sentinel-1-derived SAR vessel detections, which demonstrates that this data product exists and can potentially help reduce your preprocessing burden.

---

27. Your AIS preprocessing

You will probably need:

Raw AIS
↓
Remove invalid coordinates
↓
Sort by vessel/time
↓
Remove duplicates
↓
Handle missing values
↓
Calculate temporal gaps
↓
Interpolate only where scientifically justified
↓
Construct trajectories
↓
Calculate derived movement features
↓
Construct graph

Be especially careful with interpolation.

You don't want your preprocessing to accidentally "fill in" the exact dark-vessel gaps you're trying to detect.

---

28. Do NOT blindly interpolate AIS gaps

This is a subtle but important issue.

Suppose:

10:00 AIS
10:05 AIS
10:30 AIS

That 25-minute gap could be important.

If you automatically interpolate:

10:10
10:15
10:20
10:25

you may erase the anomaly your model should detect.

So define explicitly:

> Which gaps are retained as missing?
>
> Which gaps are interpolated?
>
> Why?

---

29. Temporal alignment

AIS is continuous-ish telemetry.

SAR is a discrete snapshot.

Therefore:

AIS: 10:00
AIS: 10:01
AIS: 10:02
AIS: 10:03
...
SAR: 10:05

You need a method for estimating:

> Where should vessel X be at SAR acquisition time?

Possible approaches include:

- nearest temporal AIS record
- interpolation
- trajectory prediction

- motion extrapolation

This is central to your thesis.

---

30. Spatial alignment

A SAR detection may not perfectly coincide with an AIS point.

You need to account for:

- positioning uncertainty
- SAR geolocation
- vessel motion between AIS transmission and satellite acquisition
- AIS update intervals

Your model should therefore probably compare **spatiotemporal proximity**, not exact coordinate equality.

---

31. Matching candidates

For each SAR detection:

SAR detection
↓
Retrieve AIS candidates
↓
Spatial filter
↓
Temporal filter
↓
Trajectory compatibility
↓
Feature similarity
↓
Cross-modal model
↓
Match / discrepancy / ambiguous

This is an excellent candidate architecture for your methodology chapter.

---

32. What your model output should look like

Don't make the output simply:

DARK = YES/NO

You can make it more informative:

SAR Detection ID
↓
Best AIS Candidate
↓
Matching Score
↓
Temporal Difference
↓
Spatial Difference
↓
Consistency Score
↓
Classification

Example:

SAR-001

Best AIS candidate: MMSI XXXXX
Spatial difference: 0.42 km
Temporal difference: 3 min
Alignment score: 0.91

Result: MATCHED

or:

SAR-002

Best AIS candidate: None

AIS coverage gap: 21 min

Result: AIS DISCREPANCY
Confidence: 0.87

Your POC can then visualize exactly this.

---

33. Your POC

Your dean made this very clear:

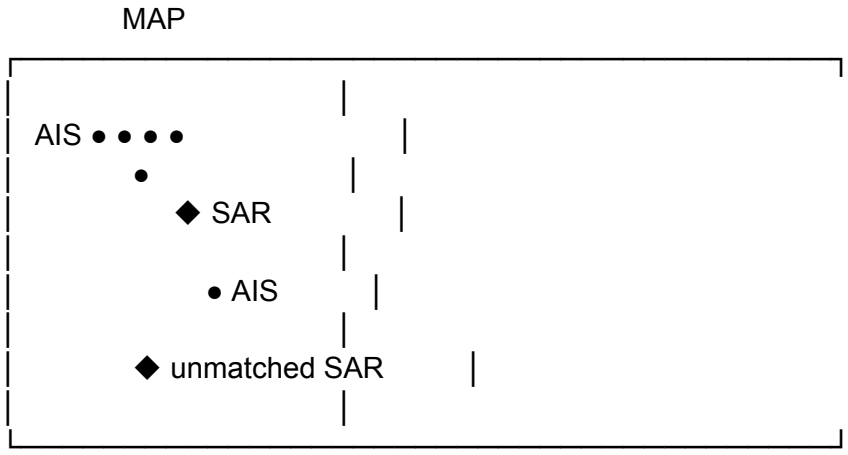
> POC required.

But:

> **POC is not the novelty.**

Your POC should demonstrate the algorithm.

A good workflow:



Click detection:

SAR Detection #038

Time:
2026-07-13 09:42 UTC

Nearest AIS:

None

Spatial search radius:
X km

Temporal window:
±X minutes

Cross-modal score:
0.83

Classification:
AIS-SAR DISCREPANCY

---

34. POC technology

You don't need to overcomplicate this.

Potential stack:

Backend

- Python
- FastAPI/Flask

ML

- PyTorch
- PyTorch Geometric or DGL

Data

- Pandas
- GeoPandas
- Rasterio
- Shapely

Map

- Leaflet
- Mapbox
- OpenLayers

Database

- PostgreSQL + PostGIS

You don't need every technology above. Choose according to what your team already knows.

---

35. What NOT to put in the POC

Don't turn it into:

> "Philippine Coast Guard Command Center 2.0"

You don't need:

- user management
- complicated dashboards
- vessel registration
- messaging
- incident management
- prediction of illegal dumping
- nationwide monitoring
- real-time satellite ingestion

Your POC only needs to demonstrate:

data → algorithm → detection → visualization

---

36. Ethics and legal considerations

Your dean already gave the right direction.

For publicly available data, you still need to document:

- source
- terms of use
- attribution
- licensing
- data retention
- responsible use

GFW currently states that its APIs are for non-commercial use and requires users to agree to terms and provide attribution in publications.

Copernicus provides public Sentinel-1 data access through the Data Space Ecosystem, though access conditions can vary by product/service.

And importantly:

Do not claim operational intelligence capability.

Your paper should say:

> "research prototype"

rather than:

> "system for identifying illegal vessels."

---

37. If you obtain PCG/PPA data later

Then this becomes a different situation.

You may need:

- formal request
- data-use agreement
- institutional authorization
- security requirements
- research ethics review depending on the data
- restrictions on publication

Especially if vessel identities, operational information, or restricted datasets are involved.

Your thesis can remain fully public-data-based if access becomes difficult.

---

38. Your research scope should explicitly exclude

I strongly recommend putting a **Scope and Delimitation** subsection that says the study does NOT attempt to:

- determine whether a vessel committed illegal dumping
- identify the perpetrator of an environmental crime
- establish legal liability
- replace Philippine Coast Guard investigation
- provide nationwide maritime surveillance
- guarantee real-time detection
- identify every vessel in Batangas Bay
- infer criminal intent

This protects your thesis from overclaiming.

---

39. Your actual contribution

Your contribution should be something like:

> **A cross-modal spatiotemporal alignment mechanism that integrates AIS trajectory information and SAR vessel observations to detect vessel-level inconsistencies while reducing false positives in congested maritime environments.**

Then your secondary contributions can include:

Dataset contribution

A curated, temporally/geographically aligned AIS–SAR dataset for your selected Batangas study area.

Method contribution

Your cross-modal graph/alignment architecture.

Experimental contribution

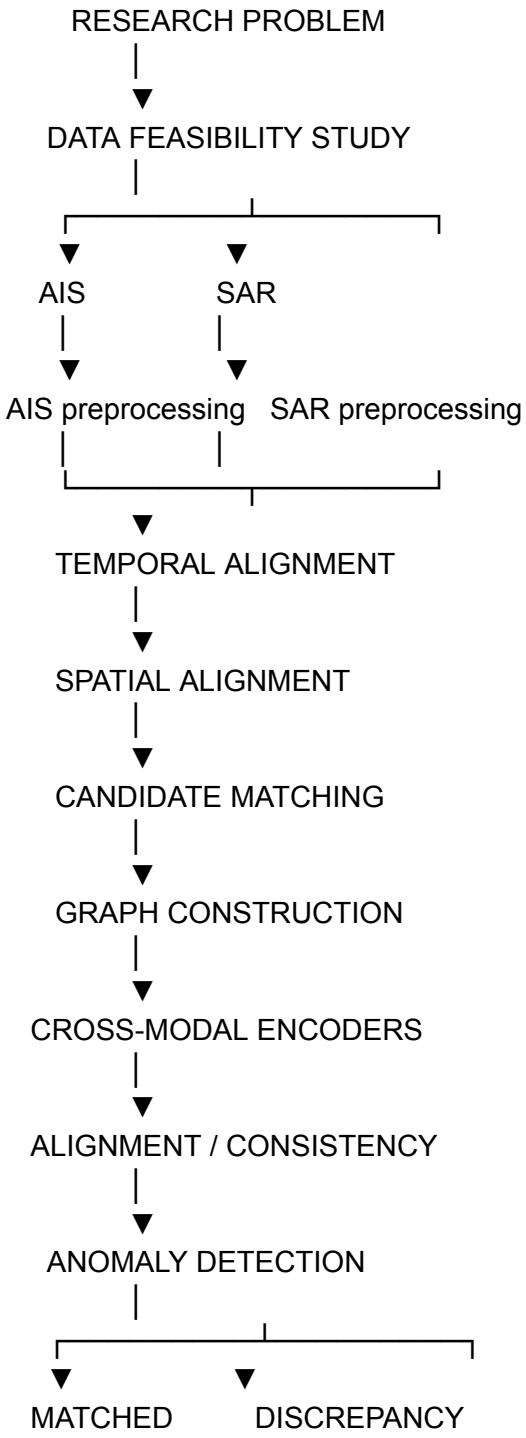
Evaluation under different congestion levels and AIS-gap conditions.

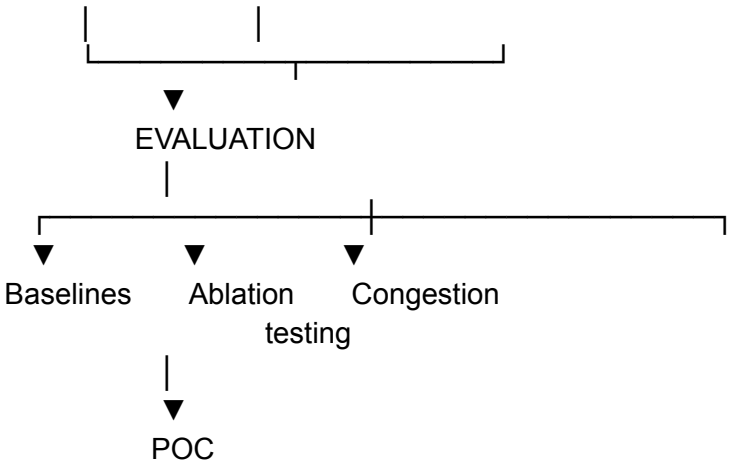
Practical contribution

A functional POC demonstrating the detected discrepancies.

40. Your thesis workflow

I would structure the entire project like this:





---

41. Experiments you should plan from the beginning

I would make a master experiment table.

Experiment	Purpose
Rule-based matching	Conventional benchmark
AIS-only	Test value of AIS alone
SAR-only	Test value of SAR alone
Proposed model	Main method
Without alignment	Test proposed alignment
Without consistency loss	Test proposed loss
Low congestion	Baseline environment
Medium congestion	Moderate complexity
High congestion	Real challenge
Normal loitering	False-positive test
AIS gap simulation	Robustness to missing AIS

Unseen vessels	Generalization
Unseen time period	Temporal generalization

That is already enough for a very solid undergraduate experimental study.

---

42. AIS-gap simulation could be very useful

A particularly useful experiment would be to deliberately create artificial AIS gaps.

For example:

Original AIS:

●—●—●—●—●—●—●—●

Simulated gap:

●—●—●　　●—●—●

Then ask:

> Can the model correctly identify when SAR observations occur during the artificial AIS gap?

This gives you controllable experimental cases.

But be careful:

simulated gaps are not equivalent to true dark vessels.

They should be reported separately from naturally occurring discrepancies.

---

43. Your model should probably output confidence

Instead of:

Dark = 1

use:

P(discrepancy | AIS, SAR) = 0.87

Then determine classification threshold on the validation set.

This gives you the ability to generate:

- precision-recall curves
- threshold analysis
- confusion matrix
- calibration analysis, if appropriate

---

44. Your confusion matrix

You should probably have something like:

	Predicted Matched	**Predicted Discrepancy**
Actually Matched	TN	FP
Actually Discrepancy	FN	TP

Then investigate specifically:

False positives

Why did your model say "dark" when it wasn't?

Possibilities:

- crowded harbor
- nearby vessel confusion
- SAR artifact
- AIS timing mismatch
- AIS receiver gap

This analysis can be just as valuable as the F1 score.

---

45. Error analysis should be a thesis subsection

Don't stop at:

> Model achieved 91% F1.

Investigate failures.

For example:

False Positive #17

SAR detects vessel.

AIS exists.

Model fails because two vessels are within 200 m.

That tells you:

> cross-modal identity resolution in dense traffic remains challenging.

That's much more academically meaningful.

---

46. Your research questions should connect directly to experiments

A strong structure would be:

RQ1

How accurately can AIS and SAR observations be aligned?

→ matching accuracy

RQ2

How accurately can the proposed model detect AIS–SAR discrepancies?

→ precision, recall, F1, FPR

RQ3

Does cross-modal alignment improve detection compared with existing baselines?

→ baseline comparison

RQ4

Does the proposed consistency component reduce false positives in congested maritime environments?

→ ablation + congestion experiment

RQ5

What computational cost is required?

→ inference time + memory/parameters

This is much cleaner than having 10 unrelated research questions.

---

47. Your Chapter 2 / RRL needs specific categories

Don't collect papers randomly.

Build your RRL around these areas:

A. AIS

- AIS architecture
- AIS limitations
- AIS transmission gaps
- AIS anomaly detection

B. SAR maritime observation

- Sentinel-1
- SAR vessel detection
- SAR ship detection
- SAR preprocessing

C. AIS–SAR fusion

- vessel matching
- multimodal fusion
- AIS/SAR correspondence

D. Spatiotemporal modeling

- trajectory modeling
- temporal graphs
- spatial graphs
- GNNs

E. Anomaly detection

- graph autoencoders
- reconstruction-based anomaly detection
- contrastive learning

F. Cross-modal learning

- multimodal representation learning
- contrastive alignment
- consistency learning

G. Maritime congestion

- port traffic
- vessel density
- loitering
- vessel behavior

H. Research gap

Then your gap should lead directly to:

existing methods → limitation → your mechanism

---

48. Your research gap must NOT be

❌ "Nobody has used AI for ships."

❌ "Nobody has combined AIS and SAR."

Those are almost certainly indefensible.

Your gap should be narrower:

> Existing studies have demonstrated AIS–SAR fusion and maritime anomaly detection, but there remains a research opportunity to investigate a **cross-modal spatiotemporal graph alignment mechanism specifically designed to distinguish true AIS–SAR discrepancies from normal vessel behavior in highly congested maritime environments.**

Then verify this carefully against the literature you actually review.

---

49. A major warning about your original "contrastive loss"

Don't include:

> "contrastive learning"

just because it sounds novel.

You need to answer:

What are the positive pairs?

Example:

AIS vessel A ↔ SAR detection A

What are the negative pairs?

Example:

AIS vessel A ↔ SAR detection B

How do you construct negatives?

This matters enormously.

Especially in a congested port, random negatives are too easy.

You may need **hard negatives**:

> nearby vessels with similar speed and heading.

That's actually highly relevant to your problem.

---

50. Hard-negative sampling could strengthen your contribution

Imagine:

SAR vessel
↓
AIS A = 0.3 km away
AIS B = 0.35 km away
AIS C = 0.4 km away

A simple nearest-neighbor algorithm may choose incorrectly.

Your model could learn:

> spatial proximity + temporal proximity + trajectory consistency + contextual information.

That is a much more compelling reason to use a cross-modal model.

---

51. Computational feasibility

You have to decide:

How large is the graph?

If Batangas has thousands of observations, a fully connected graph becomes expensive.

You might instead construct:

- spatial neighborhood graph
- temporal edges
- K-nearest neighbors
- candidate-only bipartite cross-modal edges

That keeps the problem computationally manageable.

---

52. Your thesis should have a clear computational pipeline

Something like:

Raw Data
↓
Cleaning
↓
Feature Engineering
↓
Temporal Synchronization
↓
Candidate Generation
↓
Graph Construction
↓
AIS Encoder
↓
SAR Encoder
↓
Cross-Modal Alignment
↓
Consistency / Reconstruction
↓
Anomaly Score
↓
Threshold
↓

Evaluation

This should eventually become one of your major thesis diagrams.

---

53. Reproducibility

Document:

- dataset versions
- retrieval dates
- API versions
- preprocessing parameters
- geographic bounds
- temporal bounds
- train/validation/test split
- random seeds
- model hyperparameters
- hardware
- software versions

GFW's current API documentation explicitly tracks dataset versions, and its documentation recommends using versioned/latest dataset identifiers appropriately.

This is especially important because external datasets can change.

---

54. Hardware requirements

Before selecting your architecture, determine:

- CPU
- GPU
- RAM
- storage
- training time

Raw SAR imagery can become large very quickly.

Your team should estimate:

Number of scenes
×
scene size
×
number of chips
×
storage per chip

before downloading everything.

---

55. Your biggest scope trap

Do NOT accidentally turn this into:

> **"Real-time AI maritime surveillance for the entire Batangas Bay."**

That's too broad.

Your actual thesis can be:

> **A research prototype evaluating AIS–SAR discrepancy detection within a defined portion of Batangas Bay over selected historical observation periods.**

That's completely reasonable.

---

56. Your final success criteria

Before saying the thesis is successful, define:

Data

✅ sufficient overlapping AIS/SAR observations

Model

✅ proposed cross-modal model trains successfully

Matching

✅ reliable AIS–SAR correspondence

Detection

✅ measurable anomaly/discrepancy performance

Congestion

✅ false positives analyzed under dense traffic

Ablation

✅ contribution of proposed component measured

Generalization

✅ test set separated from training

Efficiency

✅ inference cost measured

POC

✅ algorithm demonstrated visually

Reproducibility

✅ methodology documented

---

57. What your three-person team can divide into

Since you're a team of three, a practical division is:

Member 1 — Data & GIS

AIS acquisition
SAR acquisition
spatial/temporal alignment

preprocessing
dataset construction

Member 2 — Model

Graph construction
GNN
cross-modal fusion
loss functions
training

Member 3 — Evaluation & POC

baselines
experiments
metrics
ablation
error analysis
visualization/POC

But everyone should understand the complete pipeline. Don't let one person become the only person who understands the model.

---

58. What you should produce before coding

This is important.

Before seriously implementing the model, your team should have these documents:

1. Problem Definition

One-page precise definition of:

> What is an AIS–SAR discrepancy?

2. Data Specification

Exactly:

> source → fields → time → geographic region → format

3. Labeling Protocol

Exactly:

> matched / discrepancy / ambiguous

4. Data Split Protocol

Exactly:

> train / validation / test

5. Baseline Specification

Exactly which methods you're comparing.

6. Proposed Model Diagram

Exactly where your novelty lives.

7. Experiment Matrix

Exactly how you'll prove the contribution.

8. POC Specification

Exactly what the demonstrator must show.

Doing these first will save you a LOT of rewriting later.

---

59. Your thesis in one sentence

Your entire project should ultimately be understandable as:

> **We develop a cross-modal spatiotemporal graph framework that compares AIS vessel trajectories with SAR-detected vessels to identify AIS–SAR discrepancies, while reducing false positives caused by temporal gaps, nearby vessels, and normal congestion in a selected portion of Batangas Bay.**

That's your north star.

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60. The most important things to settle NOW

Before your team starts building the fancy GNN, I would put these **seven items at the top of your immediate checklist**:

Priority	Question you must answer
🔴 1	**Can we actually obtain temporally/geographically overlapping AIS + SAR data for Batangas Bay?**
🔴 2	**Will we use raw Sentinel-1 images or already-detected SAR vessels?**
🔴 3	**What exactly constitutes a "dark-vessel/AIS discrepancy" label?**
🔴 4	**What spatial and temporal tolerance defines a valid AIS–SAR match?**
🔴 5	**What exactly is novel about our cross-modal alignment mechanism?**
🟠 6	**What baselines and ablation experiments prove that novelty?**
🟠 7	**How will we prevent vessel/time leakage between training and testing?**

And honestly, **#1–#4 should be settled before #5–#7**. There is little value in perfecting the architecture if the data and ground truth cannot support the research question.

One encouraging point: the underlying data ecosystem is real. Sentinel-1 data are openly available through Copernicus, while GFW currently provides AIS services and Sentinel-1-derived SAR vessel detections, including AIS-matched/unmatched information.

Your thesis's core structure should therefore be:

AIS + SAR → temporal/spatial alignment → graph construction → cross-modal alignment → discrepancy score → congestion-aware evaluation → POC

—not:

AIS + SAR → illegal dumping detection.

That is the refinement I would carry consistently through your **Topic Abstract Form, Chapter 1, RRL, conceptual framework, methodology, experiments, and POC**.